Southampton Solent University

Application 004018

Section A: Researcher Details

Date application started:
Sun 8 February 2026 at 11:32

First name:
Partha Pratim

Last name:
Mazumder

Email:
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Research project title:
Adaptive Multi-LLM Agentic RAG with Dynamic Task Routing and Retrieval Planning
Last updated:
18/03/2026

Department:
Science and Engineering

Course:
MSc Applied AI and Data Science

Applying as:
Student

Proposed project start date
13/02/2026

Supervisor

Name	Email
Bacha Rehman	bacha.rehman@solent.ac.uk

Section B: General Questions

Are you dealing with human participants?
No

Is your research going to involve vulnerable groups in any way?
No

Will the study require the cooperation of a gatekeeper?
No

Are you dealing with Personal or Sensitive topics?
No

Will there be audio, video or photographic recording of participants?
No

Will the project involve any risk to the participants' health?
No

Could the study induce psychological stress or anxiety, or cause harm or negative consequences beyond the risks encountered in normal life to participants or researchers?
No

Does your research involve the storage of records on a computer, electronic transmissions, or visits to websites, which are associated with terrorist or extreme groups or other security sensitive material?
No

Will the project involve the development for export of 'controlled' goods regulated by the Export Control Organisation (ECO)?
No

Will the project involve financial inducement offered to participants other than reasonable expenses and compensation for time?
No

Will this study be submitted for ethical review to an external organisation?
No

Will any aspect of your research take place overseas?
No

Will the study involve use of deception?
No

Does your project involve using children as your participant population? If 'yes', for children under the age of 18, their own consent (where possible) and parental / guardian consent is required this must be written consent).
No

Are there any other ethical issues or risks of harm raised by the study that have not been covered by previous questions?
No

Field specific questions

Are you conducting research in the field of Psychology?
No

Are you conducting research in the field of Health, Exercise and Sport Science?
No

Data collection

Please be aware:

- If undertaking questionnaire based research you are required to have an ID card (your campus card will suffice) on show at all times.
- If collecting your data off campus you will be required to work in pairs (to maintain a safe environment) as well as informing one of your peers of your leaving time, location of destination and expected return time.

By ticking yes, you are agreeing to undertake this task.
Yes

Section C: Data Management and storage

Data storage

Where will data be stored during your research?
University Onedrive Account

How will data be stored after your research concludes, and for how long?
University Onedrive Account for 6 months

Data anonymisation

Will you anonymise data in storage after conclusion of research?
Yes

Section D: Participant Information, Informed Consent, and Right To Withdraw

Will all the participants be fully informed BEFORE the project begins why the study is being conducted and what their participation will involve?
- not entered -

Will every participant be asked to give written consent (or assent if working with minors) to participating in the study, before it begins?
- not entered -

Will all participants be fully informed about what data will be collected, and what will be done with this data during and after the study?
- not entered -

Participants will be informed that study data may be used in future presentations and/or publications
- not entered -

Are you dealing with Personal or Sensitive topics?
No

Participants' anonymity will be maintained in this study
- not entered -

Participants' confidentiality will be maintained in this study
- not entered -

Participants will be provided with adequate information to understand what the purpose of the study was
- not entered -

Participants will be provided with details of how to contact you should they wish to do so
- not entered -

Participants will be offered access to a summary of the results
- not entered -

Participants will be provided with details on how to find external sources of information and/or support relating to issues in the study? (e.g. Wellbeing Hub, Samaritans)
- not entered -

Participant information

Sampling procedure (e.g. convenience, purposive, theoretical)
- not entered -

Characteristics of the participants you wish to use (e.g. gender, age)
- not entered -

Participant inclusion and exclusion criteria (e.g. inclusion - must play competitive sport; exclusion: lower limb injury in the last 6 months)
- not entered -

Data anonymisation

Will every participant understand that they have the right not to take part at any time, omit questions, and/or withdraw themselves and their data from the study if they wish?
- not entered -

Please justify why participants cannot omit questions/topics
- not entered -

Section E: Study Details and Additional Risks Information

Study details

What is the purpose of the project? (up to 500 words)

The purpose of this research is to investigate and improve how conversational AI systems handle complex user queries. Many existing AI assistants rely on a uniform processing strategy, applying computationally intensive techniques such as external retrieval or multi-step agentic reasoning to all queries regardless of their complexity. This approach can lead to inefficiencies, slower response times, and an increased risk of inaccurate outputs.

This project proposes an adaptive conversational framework inspired by human problem-solving, where different strategies are applied depending on the nature of the task. The system is designed to first classify the type of user query—such as factual, analytical, or procedural—and then dynamically route it to the most appropriate language model or tool. Advanced retrieval and multi-step planning mechanisms are invoked only when necessary.

By aligning computational resources with task requirements, the proposed approach aims to improve accuracy, efficiency, and cost-effectiveness. Ultimately, this research contributes toward the development of more scalable, reliable, and intelligent conversational systems.

Outline of the project (up to 500 words)

This project involves the design and evaluation of a novel adaptive AI system architecture using frameworks such as LangGraph for query understanding, dynamic task routing, and delegation to pretrained large language models (LLMs).
The system will be evaluated computationally using publicly available benchmark datasets (e.g., HotpotQA or similar), with no human participants involved. Performance will be compared against non-adaptive approaches based on accuracy, response time, and computational cost.

What methods will you use to collect your data?
Data for training and evaluating the system will be sourced exclusively from publicly available, benchmark datasets (HotpotQA). These datasets are standard in AI research and contain de-identified, anonymized question-answer pairs. No primary data collection from human subjects, social media, surveys, or interviews will be conducted. All interactions are between the AI system and these static datasets. No questionnaires, clinical scales, or licensed instruments are involved.

During your data collection will supervision or assistance be required (e.g. for experiments in the physiology laboratory, therapy laboratory or teaching gym)?
No

Research

What is the desired outcome of the project? (300 words)
The desired outcome is a validated, efficient framework for adaptive AI reasoning. Specifically, the project aims to produce:

1) A functional prototype of an adaptive multi-LLM system with dynamic routing logic.
2) Empirical evidence demonstrating that this adaptive approach outperforms standard single-model or always-on agentic systems in key metrics such as answer accuracy (F1/Exact Match scores), reduced computational latency, and lower operational cost per query.
3) A set of design principles and performance benchmarks for when to use retrieval, specialized models, or agentic planning.

The findings will be disseminated through a Master's thesis and potentially a scholarly publication, contributing to the academic field of efficient AI system design. The ultimate goal is to provide a blueprint for building more responsive and resource-conscious conversational AI assistants.

Competence (i.e. Explain how you are sufficiently competent to undertake any data collection and, or training schedules)
I, Partha Pratim Mazumder, am a postgraduate research student in Applied AI and Data Science at Southampton Solent University. I possess strong technical competency in Python programming, machine learning concepts, and the use of large language model APIs, which are essential for this project. I am proficient in using relevant software libraries and development frameworks. My academic coursework and independent study have provided a solid theoretical foundation in natural language processing and AI system architecture. Furthermore, I will be under the direct supervision of Dr. Bacha Rehman, an experienced academic who will provide guidance on research methodology, technical implementation, and ethical compliance throughout the project duration. Regular supervisory meetings are scheduled to monitor progress and address any challenges.

Other risks and ethical issues

Do you foresee any risks to yourself and/or other investigators in conducting this study?
No

Do you foresee any risks to the University in conducting this study?
No

Are there ANY OTHER ethical issues not covered elsewhere in this submission that should be brought to the Ethics Committee's attention?
No

Section F: External Collaborators and Third Parties

Will you be working with any external collaborators or third parties
No

Section G: Overseas Research

Will any of your project take place offsite?
No

Will you or any Southampton Solent University researchers be travelling outside the UK?
No

Official notes

- not entered -